

ZISHWILI MAYIBONGWE CYPRIAN

Ngwenya, Oshoek, Eswatini

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Heavy Duty Driver's License | Police Clearance Certificate (no criminal record)

PROFESSIONAL SUMMARY

Registered Medical Laboratory Technologist with a Bachelor of Science in Medical Laboratory Sciences (Second Class Upper, CGPA 4.36/5.50). Six months of hands-on clinical placement in an ISO 15189-accredited, high-throughput diagnostic laboratory (Cerba Lancet Eswatini & Eswatini Hospital Laboratory Services). Experienced across all major laboratory disciplines: haematology, clinical chemistry, microbiology (routine and TB), immunology & serology, blood transfusion, parasitology, cellular pathology, and molecular biology (basics). Certified in phlebotomy, workplace safety, and first aid. Proven ability to operate a wide range of automated analysers, implement quality control (IQC), follow strict SOPs, and improve workflow efficiency. Committed to diagnostic accuracy, patient safety, and public health.

EDUCATION

Bachelor of Science in Medical Laboratory Sciences (Second Class Upper) Conferred 28 March 2026

Eswatini Medical Christian University (EMCU)

CGPA 4.36/5.50

- Full spectrum coursework: Haematology (I & II, Haempathology), Clinical Chemistry (I, II, III), Medical Microbiology (I, II, Virology & Parasitology), Immunology & Serology, Blood Transfusion Science, Cellular Pathology, Molecular Biology, Medical Genetics, Research Methods, Laboratory Management (QMS, ISO 15189, SOPs).

High School – EGCSE Certificate

Nov 2020

Ka-Boyce High School

- 7 credits including Mathematics and Physical Science.

CERTIFICATIONS & REGISTRATION

- Registered Medical Laboratory Technologist – Eswatini Medical & Dental Council (registration pending)
- Diploma in Phlebotomy (ID: 45878632) – Alison (Feb 2026)
- Diploma in Workplace Safety & Health – Alison (Aug 2025)
- Secretarial Studies & Office Administration – Alison (Aug 2025)
- EF SET English B2 (55/100) – EF Education First (Oct 2025)
- First Aid Blended Learning (80% pass) – Baphalali Eswatini Red Cross (Nov 2024)
- ISO 15189 Laboratory Quality Management – In-house training (EHLS, 2024)

LABORATORY COMPETENCIES (BY DEPARTMENT)

Haematology & Haemostasis

- Peripheral blood smear preparation, staining, and microscopic examination (manual differential)

- Complete Blood Count (CBC) analysis using automated analysers (Mindray BC-5380, Act Diff 5)
- Malaria microscopy (thick and thin films)
- INR and basic coagulation testing
- Haemoglobinopathy and haemostasis fundamentals (from Haematology II & Haem-pathology)

Clinical Chemistry

- Operation of Beckman Coulter AU480 and Roche Cobas c111/e411 analysers
- Daily Internal Quality Control (IQC), calibration drift identification and correction
- Spectrophotometry, colorimetry, alkalinity titration, HPLC principles
- Hormone assays, tumour markers, therapeutic drug monitoring (basics)
- Blood gas analysis (epoc system – CO₂, O₂)

Microbiology (General & TB)

- General: Gram staining, Ziehl-Neelsen (ZN) staining, culture media preparation, bacterial isolation (streaking), antibiotic susceptibility testing (disk diffusion), urine/stool/sputum processing, microscopy for parasites (wet prep, iodine, saline)
- TB specific: Sputum preparation for GeneXpert molecular testing inside Class II biosafety cabinet; modified AFB (Ziehl-Neelsen) staining and microscopy for treatment monitoring; rapid TB LAM urine assay; infection control for high-risk specimens

Immunology & Serology

- Rapid diagnostic tests (RDTs): HIV combo, Syphilis (RPR – qualitative and semi-quantitative), HBsAg, TB LAM
- Operation of Beckman Coulter UniCel Dxl 800 (immunoassay analyser)
- ELISA principles, agglutination reactions, complement fixation (academic foundation)

Blood Transfusion Science

- ABO and Rhesus blood grouping (manual and using QUALYS automated system)
- Cross-matching (major and minor), antibody screening
- 3-5% red cell suspension preparation
- Transfusion reaction investigation basics

Cellular Pathology & Histology

- Tissue processing (fixation, embedding, microtomy)
- Routine staining (H&E) and Papanicolaou (Pap) staining
- Fine needle aspiration cytology (FNAC) fundamentals (from ML 408 Cellular Pathology)

Molecular Biology & Medical Genetics

- DNA/RNA extraction fundamentals, PCR principles, gel electrophoresis
- Gene sequencing basics (from Medical Genetics), FISH, karyotyping
- Operation of Panther Hologic (TTI, HPV, COVID-19) and Cobas 4800 (viral load) – placement experience

Parasitology

- Microscopic identification of intestinal parasites (wet preparation, saline and iodine mounts)
- Malaria parasite detection and species identification (training from ML 410)

Laboratory Management & Quality Systems

- ISO 15189 quality management system (QMS) fundamentals
- Standard Operating Procedure (SOP) adherence and documentation

- Internal Quality Control (IQC) and external quality assessment (EQA) awareness
- Sample reception, registration, tracking, archiving using LIMS (Disa, MEDITECH)
- Non-conformance resolution (e.g., destruction record verification)
- Inventory management and reagent stock control (basic)

LABORATORY EXPERIENCE

Clinical Attaché (6-Month Rotation Across All Departments) Apr 2024 – Nov 2024
Cerba Lancet Eswatini (Apr–Jul) & Eswatini Hospital Laboratory Services (Aug–Nov)
 ISO 15189-accredited, high-throughput diagnostic laboratory

- **Sample Management & Data Entry:** Received, registered, centrifuged, aliquoted, and archived specimens using LIMS (Disa, MEDITECH) and HMIS.
- **Haematology & Blood Transfusion:** Performed CBC on Mindray BC-5380 and Act Diff 5; prepared and stained peripheral smears for manual differentials and malaria microscopy. Conducted ABO/Rh grouping (manual and QUALYS), cross-matching, and antibody screening.
- **Clinical Chemistry:** Operated Beckman Coulter AU480 and Roche Cobas c111/e411; ran daily IQC, corrected calibration drift, and prevented repeat analyses.
- **Microbiology (General & TB):** Processed urine, stool, and sputum for routine culture; performed Gram and ZN staining, media preparation, and antibiotic susceptibility testing. In TB laboratory: prepared sputum for GeneXpert inside Class II biosafety cabinet; ensured zero late/incorrect results; monitored treatment response using modified AFB staining and microscopy; performed TB LAM rapid test.
- **Immunology & Serology:** Executed RDTs for HIV, Syphilis (RPR), HBsAg, and TB LAM. Operated Beckman Coulter UniCel Dxl 800.
- **Molecular & Specialised Testing:** Assisted with viral load testing (Cobas 4800), CD4 analysis (Sysmex CyFlow Counter), HPV and TTI screening (Panther Hologic).
- **Quality, Safety & Compliance:** Strictly followed SOPs, biosafety (BSL-2), patient confidentiality, and hazardous waste disposal. Participated in non-conformance resolution.

Pre-Analytical Technician & Phlebotomist March 2026 – Present
Dimer Laboratories (Mbabane)

- Perform daily phlebotomy (20-30 patients), specimen processing, and data entry.
- Identified turnaround time delays and implemented an urgent results delivery protocol, improving patient satisfaction and workflow efficiency.

RESEARCH EXPERIENCE

Undergraduate Thesis Approved June 2025
Prevalence of Hepatitis B Virus in Pregnant Women Attending One Antenatal Care Facility in Mbabane
 EMCU, Department of Medical Laboratory Sciences | Supervisor: Ms. Mdluli Nomkhosi

- Designed a retrospective cross-sectional study (January 2021 – December 2024).
- Obtained ethical approval from Eswatini Health and Human Research Review Board (EHRRB).
- Collected and analysed data using Microsoft Excel (descriptive statistics, 95% confidence intervals).
- Key findings: 3.5% HBV prevalence, 33.3% HIV-HBV co-infection, 0% vaccination coverage.

- Produced evidence-based public health recommendations to strengthen antenatal care and prevent mother-to-child transmission.

PROFESSIONAL ATTRIBUTES

- Detail-oriented with strong documentation and reporting skills
- Analytical and critical thinking, problem-solving under pressure
- High manual dexterity (phlebotomy, fine laboratory techniques)
- Team collaboration and effective communication across disciplines
- Time management and ability to handle high-volume workloads
- Adaptable and quick learner – mastered new analysers within days
- Strong work ethic, proactive, and committed to continuous improvement
- Dedicated to quality, safety, and patient confidentiality

LANGUAGES

- **English:** Professional proficiency (EF SET B2 – 55/100)
- **Siswati:** Native

REFERENCES

1. **Ms. Therese Lusandza Mulyangote**
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All references have confirmed their willingness to provide professional recommendations. Contact details are shared with their permission.